

Chapter 8: Why Are Transformers So Successful?

Transformers are currently the most powerful neural network architecture, especially in **Natural Language Processing (NLP)**.

They are also becoming state-of-the-art in **Computer Vision** (image processing).

The success of transformers mainly depends on four key factors:

1. Attention Mechanism
2. Self-supervised Pretraining
3. Large Number of Parameters
4. Easy Parallelization

Now let's understand each one step by step.

1. Attention Mechanism (Most Important)

The core strength of transformers is the **self-attention mechanism**.

Simple Explanation:

Consider the sentence:

"The cat sat on the mat because it was tired"

Here, what does "it" refer to?

It refers to "cat".

What does the transformer do?

- It compares each word with every other word
- It identifies which words are important

Main Idea:

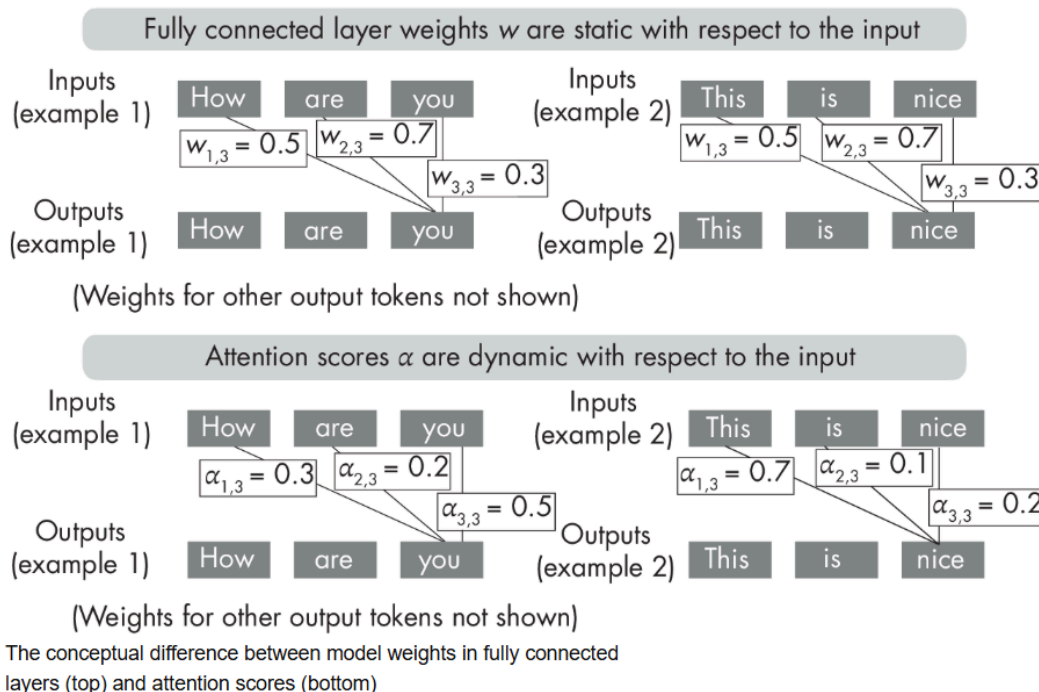
Each word gives "attention" to all other words in the sentence.

What Does Self-Attention Do?

- Each token (word) finds relationships with all other tokens
- Important words get higher weight

Example:

- "cat" → important
- "the" → less important



Important Difference (Very Important for Exams)

Fully Connected Layer:

- Weights are fixed after training

Attention Mechanism:

- Weights are dynamic (change depending on input)

This means:

The same transformer model can behave differently for different inputs.

Why Is Attention Powerful?

- It understands context
- It handles long sentences well
- It focuses on important information

That is why transformers are highly accurate.

2. Pretraining (Self-Supervised Learning)

Transformers are first trained on large datasets without labels.

What Do They Learn?

- Predict missing words
- Predict the next sentence

Example:

"I love ____ learning"

→ The model predicts "machine"

Benefits:

- Learns language structure automatically
- Can be fine-tuned for smaller tasks later

This process is called:

Pretrain → Fine-tune

3. Large Number of Parameters

Transformer models are very large.

Example:

- GPT-3 → 175 billion parameters

What Are Parameters?

Parameters represent the learning capacity of the model.

- More parameters → more knowledge can be learned
- Can capture complex patterns

Scaling Law (Important Concept)

When model size increases:

- Training loss decreases
- Performance improves

Rule:

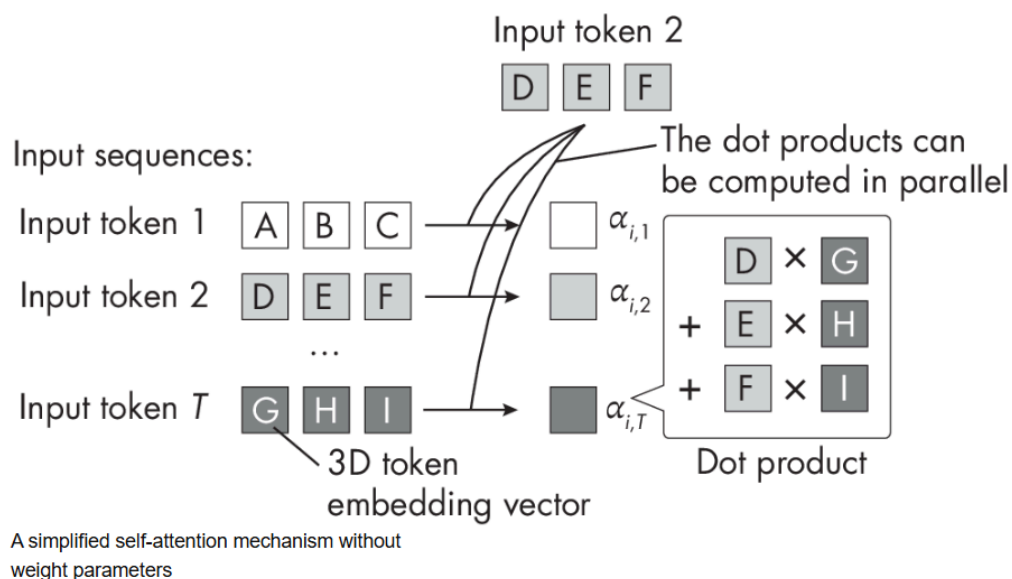
If model size doubles → training data should also double

Important Point:

Large data + large model = best performance

4. Easy Parallelization

Transformer training is fast because it supports parallel computation.



Problem with Older Models (RNN):

- Sequential processing (one step at a time)
- Slow training

What Do Transformers Do?

- Process all tokens at the same time
- Run efficiently on GPUs

Example:

In self-attention:

- All words are compared simultaneously
- Computations are independent

Therefore:

- Faster training
- Enables very large models

References

- An example of an attention mechanism in the context of image recognition: Hugo Larochelle and Geoffrey Hinton, “Learning to Combine Foveal Glimpses with a Third-Order Boltzmann Machine” (2010), <https://dl.acm.org/doi/10.5555/2997189.2997328>.
- The paper introducing the self-attention mechanism with the original transformer architecture: Ashish Vaswani et al., “Attention Is All You Need” (2017), <https://arxiv.org/abs/1706.03762>.
- Transformers can have trillions of parameters: William Fedus, Barret Zoph, and Noam Shazeer, “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity” (2021), <https://arxiv.org/abs/2101.03961>.
- Linear scaling laws suggest that training loss decreases proportionally with an increase in model size: Jared Kaplan et al., “Scaling Laws for Neural Language Models” (2020), <https://arxiv.org/abs/2001.08361>.

- Research suggests that in transformer-based language models, the training tokens should be doubled for every doubling of model size: Jordan Hoffmann et al., “Training Compute-Optimal Large Language Models” (2022), <https://arxiv.org/abs/2203.15556>.
- For more about the weights used in self-attention and cross-attention mechanisms, check out my blog post: “Understanding and Coding the Self-Attention Mechanism of Large Language Models from Scratch” at <https://sebastianraschka.com/blog/2023/self-attention-from-scratch.html>.